

Winning Back the Base

A Value-Based Customer Retention Strategy for Company A

Turning 100,000 customer records into a quantified retention program — targeting not just who will churn, but who is most valuable to keep, then proving ROI with a paid A/B pilot.

0.692

XGBoost ROC-AUC (CV-validated)

\$10.1M

net benefit / yr (per 1M subs)

1.6x

more revenue saved vs. churn-only

Executive summary

Company A loses customers faster than it can profitably replace them. We can fix that.

■ The problem

In a saturated market, ~95% of growth comes from switchers and re-acquiring a lost customer costs 5-25x more than keeping one. Churn is Company A's most expensive leak.

■ What the data says

Churn is predictable and concentrated. The strongest driver is aging handsets - devices past ~12 months churn at 54-58% vs ~42% when new - a lever Company A controls.

■ Our solution

A monthly value-based targeting engine: rank customers by Expected Value-at-Risk (P(churn) x customer value), then make proactive, persona-specific retention offers.

■ Why it works

The model is moderately predictive (AUC ~0.69) - but the value comes from value-weighted targeting, calibration, fairness and A/B validation, not from over-claiming accuracy.

What it delivers - per 1,000,000 subscribers

Best model	XGBoost · AUC 0.692
Targeting edge	captures 44% of revenue-at-risk in top 20%
Net benefit / yr	\$10.1M (1.68x ROI)
Customers saved / yr	~16,909
Our fee	\$840k/yr · 8.3% of value created

Sources: Bain & Company; Harvard Business Review; analysis of Company A dataset (n=100,000).

Market analysis: retention is the only profitable growth lever

\$344B

US wireless market (2025)

1.0%

Best-in-class monthly churn

5-25x

Re-acquire vs. retain cost

25%+

Profit lift from +5% retention

■ **The market is saturated**

Penetration exceeds 100%. Carriers grow almost entirely by stealing switchers - making every avoided cancellation worth far more than a new-customer ad.

■ **Acquisition is expensive**

Industry CAC runs into the hundreds of dollars per line; retention offers are a fraction of that, so saved customers are immediately accretive.

■ **Small churn moves, big profit**

Bain's classic finding: a 5-point retention improvement can lift profits 25-95%. Churn is the highest-leverage metric in the P&L.

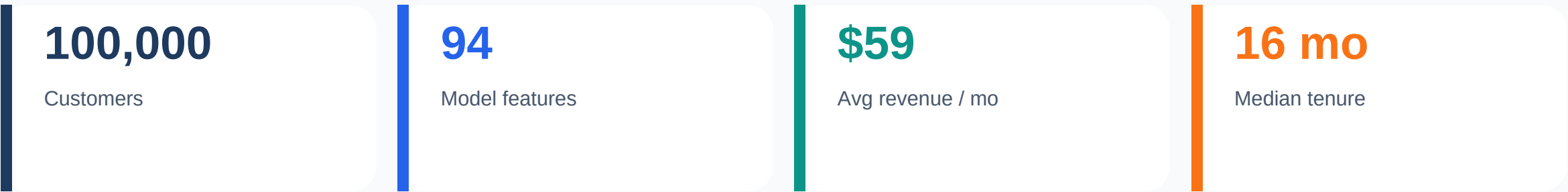
■ **Loyalty compounds**

Tenured customers spend more, cost less to serve, and refer others - so protecting the base protects future revenue too.

Sources: industry market sizing 2025; Verizon/T-Mobile investor reports; Bain & Company; Reichheld, HBR.

The client & the data

Company A is a US wireless carrier facing elevated churn. We received two linked tables covering 100,000 customers - behavioural usage and customer profile - joined on Customer_ID.



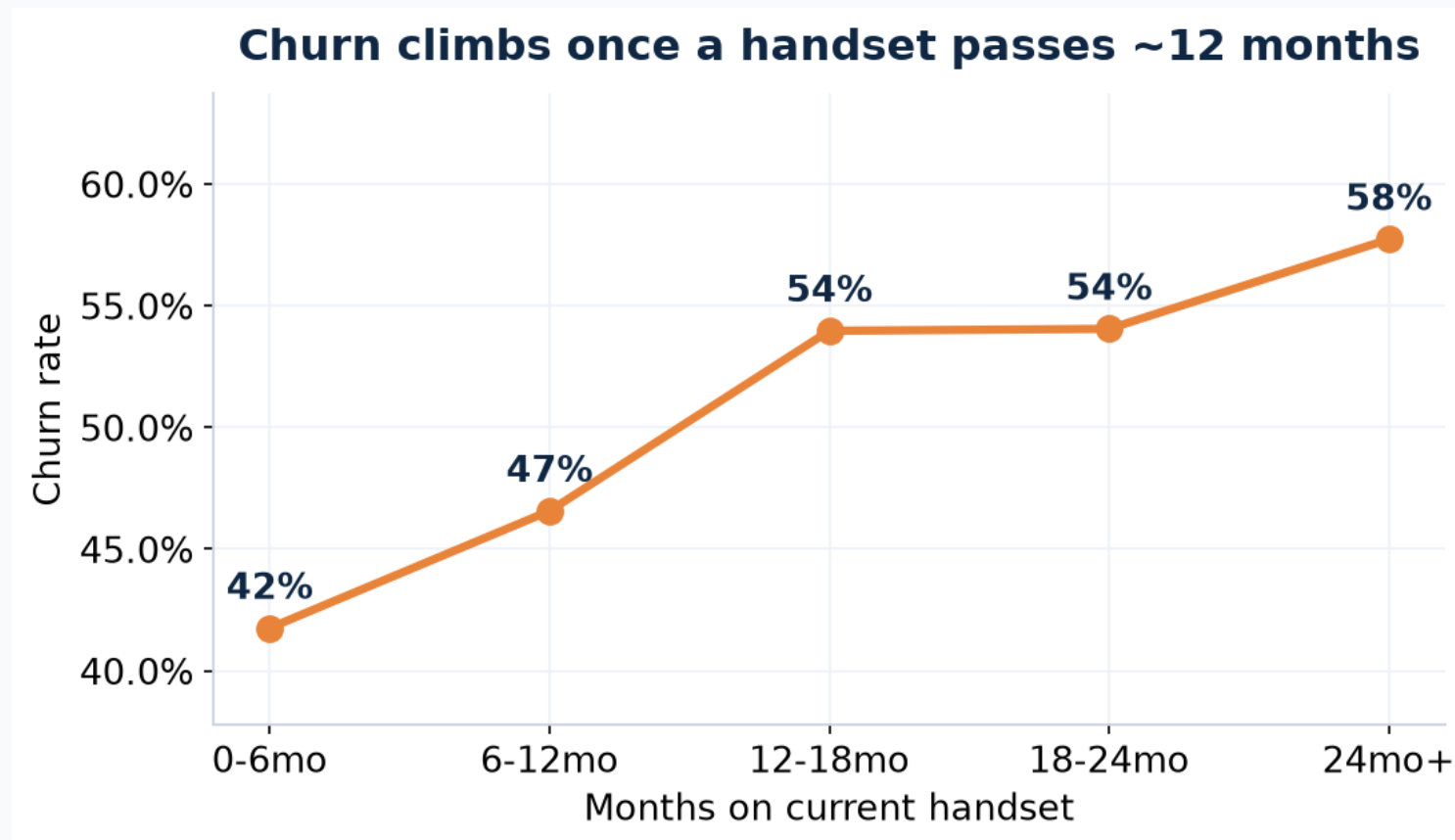
Behavioural (Record.csv)
Minutes of use, overage, revenue, call quality, recharges, and the churn label - the in-life signals of disengagement.

Profile (Client.csv)
Handset age & price, refurbished flag, web-capability, tenure - the structural drivers we can act on.

Data discipline: (1) the modelling sample is balanced ~50/50 by design (oversampled), so it is used for ranking - the real operating churn assumption (22%/yr) is applied separately in the business case; (2) we DROP 9 fields with 25-49% missingness and 4 protected attributes (ethnicity, marital status, income, credit class) - see slide 15 for the fairness test.

Source: Company A dataset (Client.csv + Record.csv), 100,000 customers.

What the data reveals: the handset-age cliff



Equipment age is the #1 signal

Time on the same handset is the strongest correlate of churn (+0.11). Churn climbs from ~47% (6-12 mo) to ~58% beyond two years.

An actionable lever

Unlike age or income, handset age is something Company A controls - via proactive upgrade & trade-in offers timed to the ~12-month cliff.

Investment buys loyalty

Higher handset price and recurring charge correlate negatively with churn - customers with newer, pricier phones stay.

Engagement matters

Declining minutes-of-use is an early-warning sign we engineered into the model as a usage-trend feature.

Source: churn rate by months-on-handset & point-biserial correlations, Company A dataset (n=100,000).

Modelling approach: rigorous, reproducible, honest

01

Clean & engineer

Drop 9 high-missing + 4 protected fields; impute, encode; engineer 7 features (usage trend, overage share, handset months...). 94 model features.

02

Benchmark + ensemble

4 model families (LogReg, RF, XGBoost, LightGBM) plus soft-voting & stacking ensembles - judged on revenue capture, not AUC alone.

03

Validate & adjust

75/25 stratified split + 5-fold CV; isotonic-checked reliability (ECE 0.7%); prior-adjusted to the 22% base rate, pilot-calibrated.

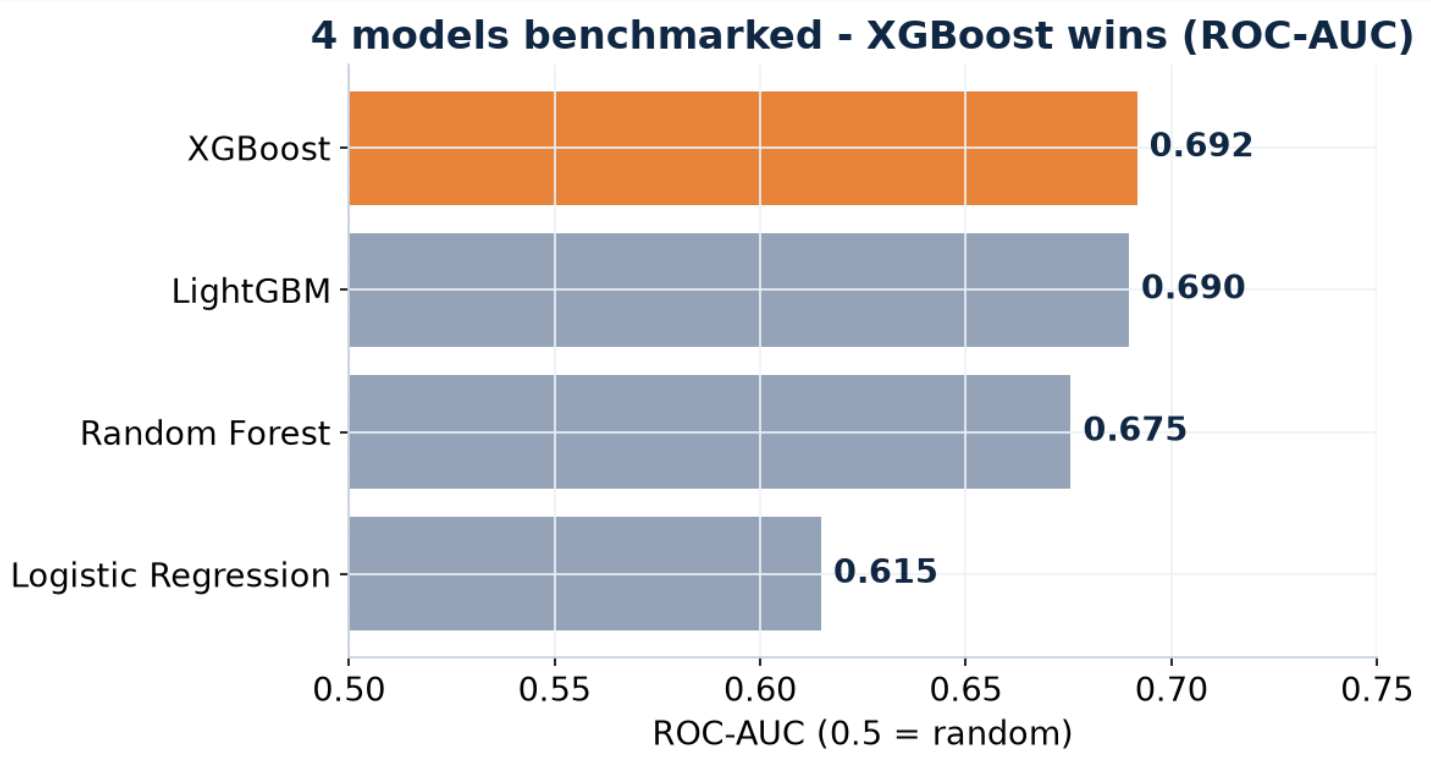
04

Rank by value

Convert to Expected Value-at-Risk; profit-curve sets the contact threshold; simulate vs random and churn-only baselines.

Cross-validation result: XGBoost ROC-AUC = 0.695 +/- 0.003 across 5 folds - confirming the model generalises and the score is stable, not luck.

Results: XGBoost wins - AUC 0.692



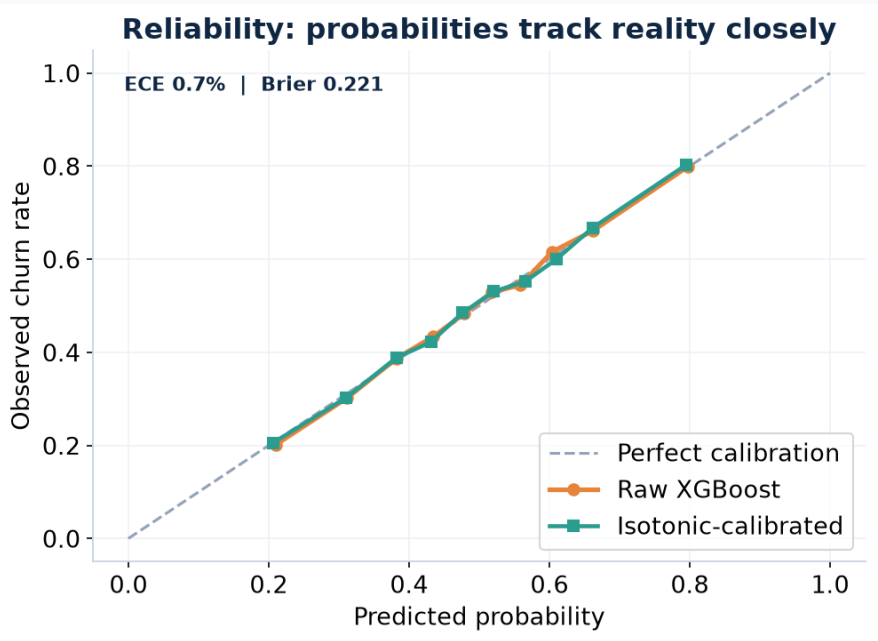
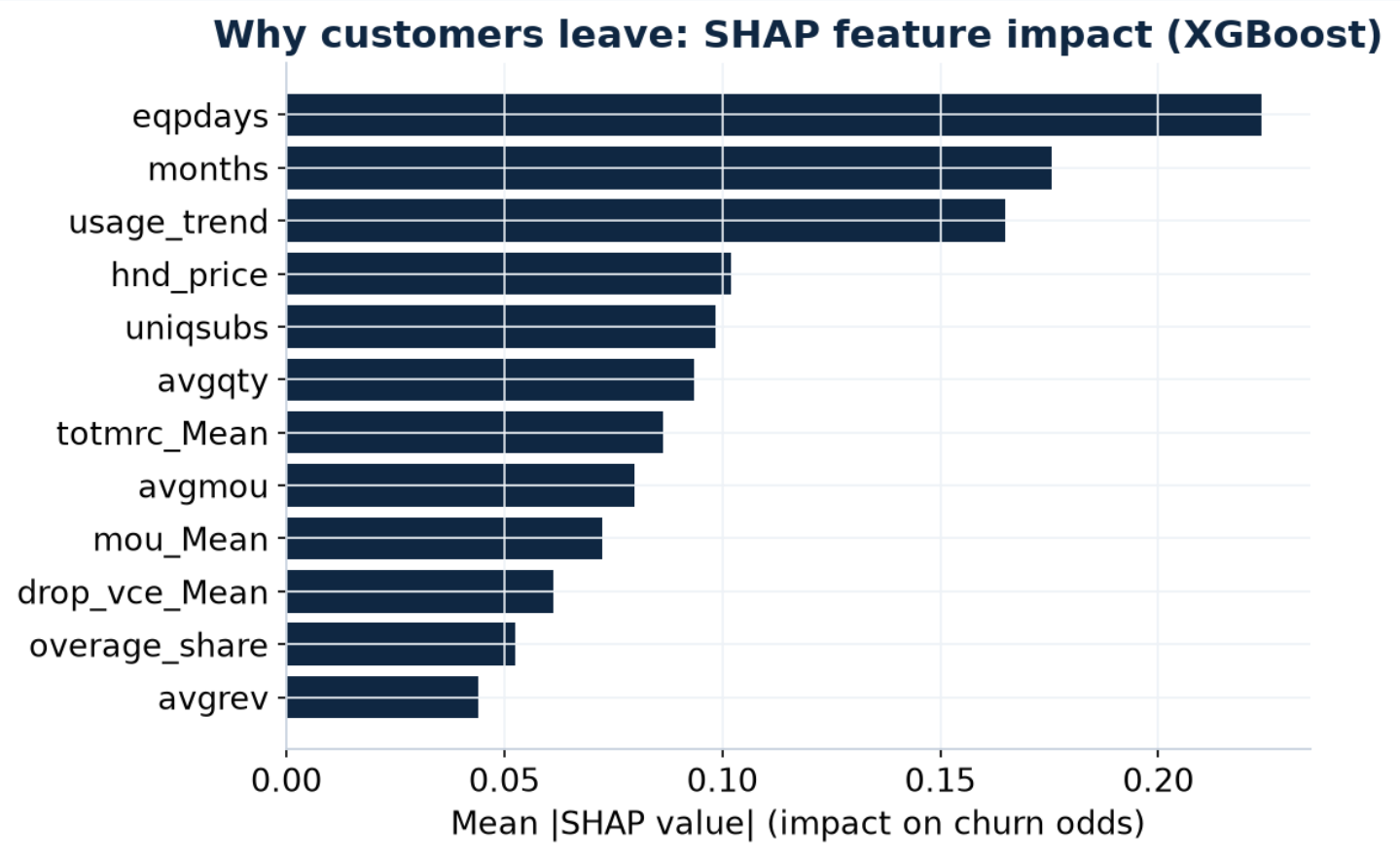
Model	AUC	Acc	Prec	Recall
Logistic Regression	0.615	0.586	0.585	0.563
Random Forest	0.675	0.619	0.612	0.630
LightGBM	0.690	0.633	0.630	0.632
XGBoost	0.692	0.634	0.631	0.631

How to read AUC: 0.5 = guessing, 1.0 = perfect. At 0.692 the model is moderately predictive - it ranks a churner above a non-churner ~69% of the time. We do not over-sell this: the payoff comes from value-weighted targeting, not raw accuracy.

Ensembles tested honestly: soft-voting & stacking reached AUC 0.6921 (+0.0005) - below our 0.003 bar - so we keep XGBoost for simpler, governable deployment.

Source: held-out test set (25% of 100,000), Company A dataset.

Why we trust the model: explainable & calibrated



- Explainable (SHAP)**

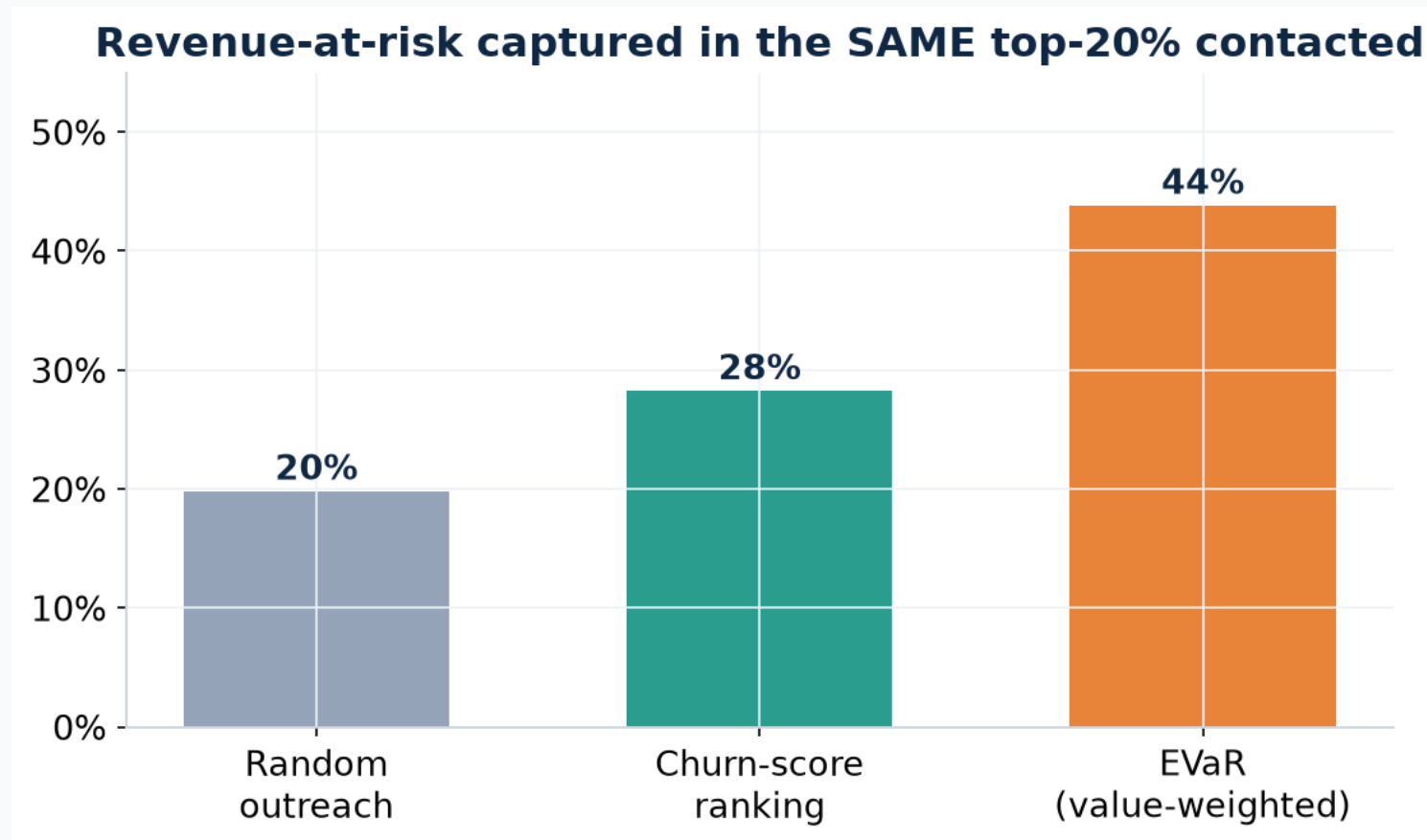
The top churn drivers are eqpdays, months and usage_trend - all behaviour Company A can act on. No protected attribute appears.
- Well-calibrated**

Reliability error is only ECE 0.7% - probabilities already track reality; isotonic confirms it. We still prior-adjust 50/50->22% for dollar scale.
- Actionable**

Because drivers are controllable, every score maps to a concrete offer.

Source: SHAP values on held-out set; reliability curve (10 bins), isotonic-checked; ECE/Brier reported.

Our differentiator: target value, not just probability



The insight

Ranking by churn probability alone wastes budget on low-value churners. Ranking by Expected Value-at-Risk protects the revenue that actually matters.

44%

of revenue-at-risk captured in top 20%

vs 28%

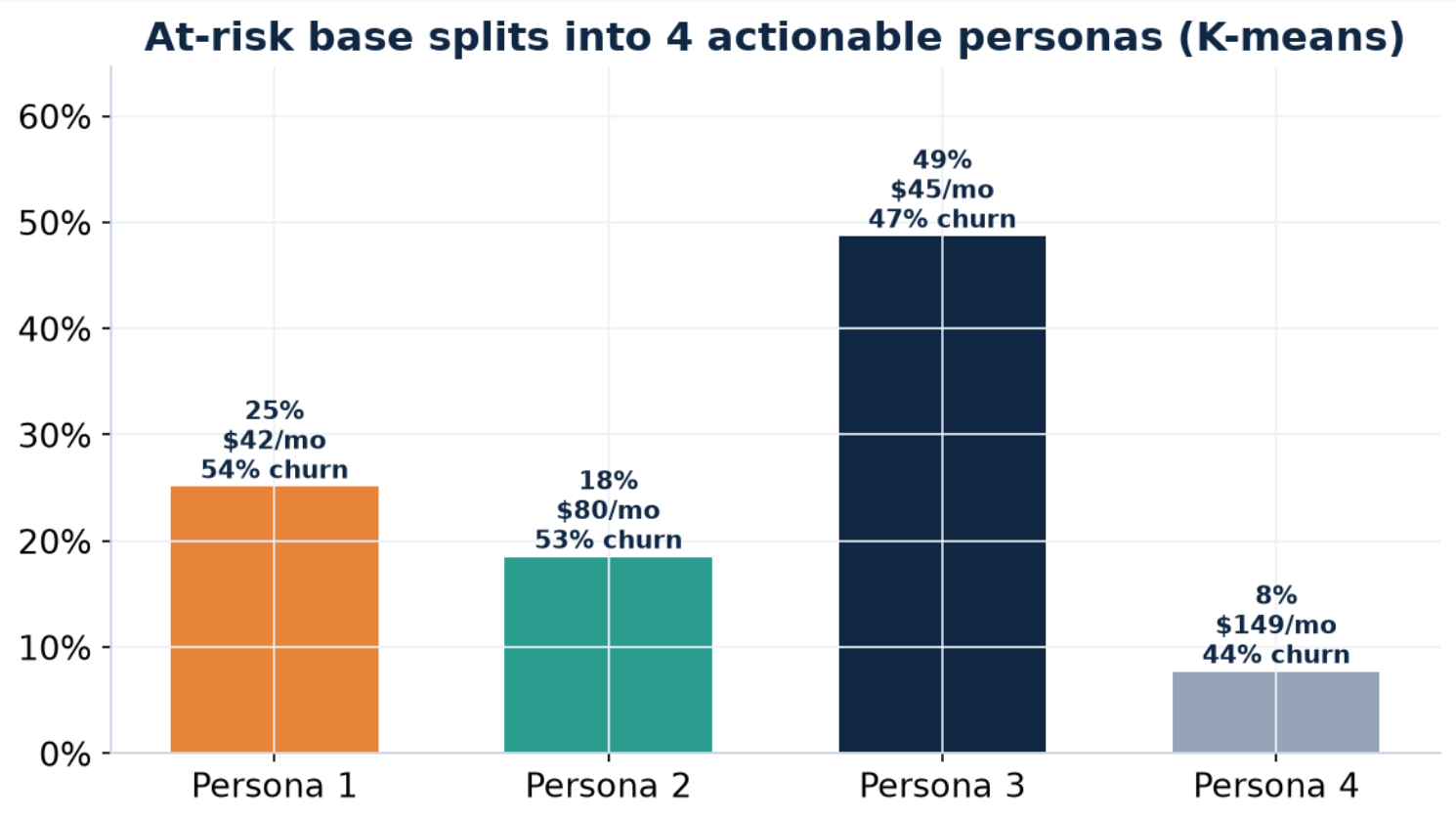
churn-only ranking

1.6x more revenue protected

Same campaign budget, same 20% of customers contacted - but we shield 1.6x the revenue by leading with value.

Source: gains simulation on held-out test set; EVaR = calibrated $P(\text{churn}) \times \text{annualised customer revenue}$.

Who to target: four data-derived retention personas



Upgrade-Ready Switchers

25% · \$42/mo · 54% churn

Aging device, modest spend
-> Proactive upgrade / trade-in

Bill-Shock Frustrated

18% · \$80/mo · 53% churn

High bill, heavy overage
-> Right-size plan + autopay discount

Low-Engagement Flight Risks

49% · \$45/mo · 47% churn

Newer, mid-spend, largest group
-> Onboarding nudges + loyalty perks

High-Value Protect-and-Hold

8% · \$149/mo · 44% churn

Top spend & usage
-> White-glove retention, protect margin

Source: K-means (k=4) on standardised behavioural features of the customer base, Company A dataset.

From segment to action: the treatment playbook

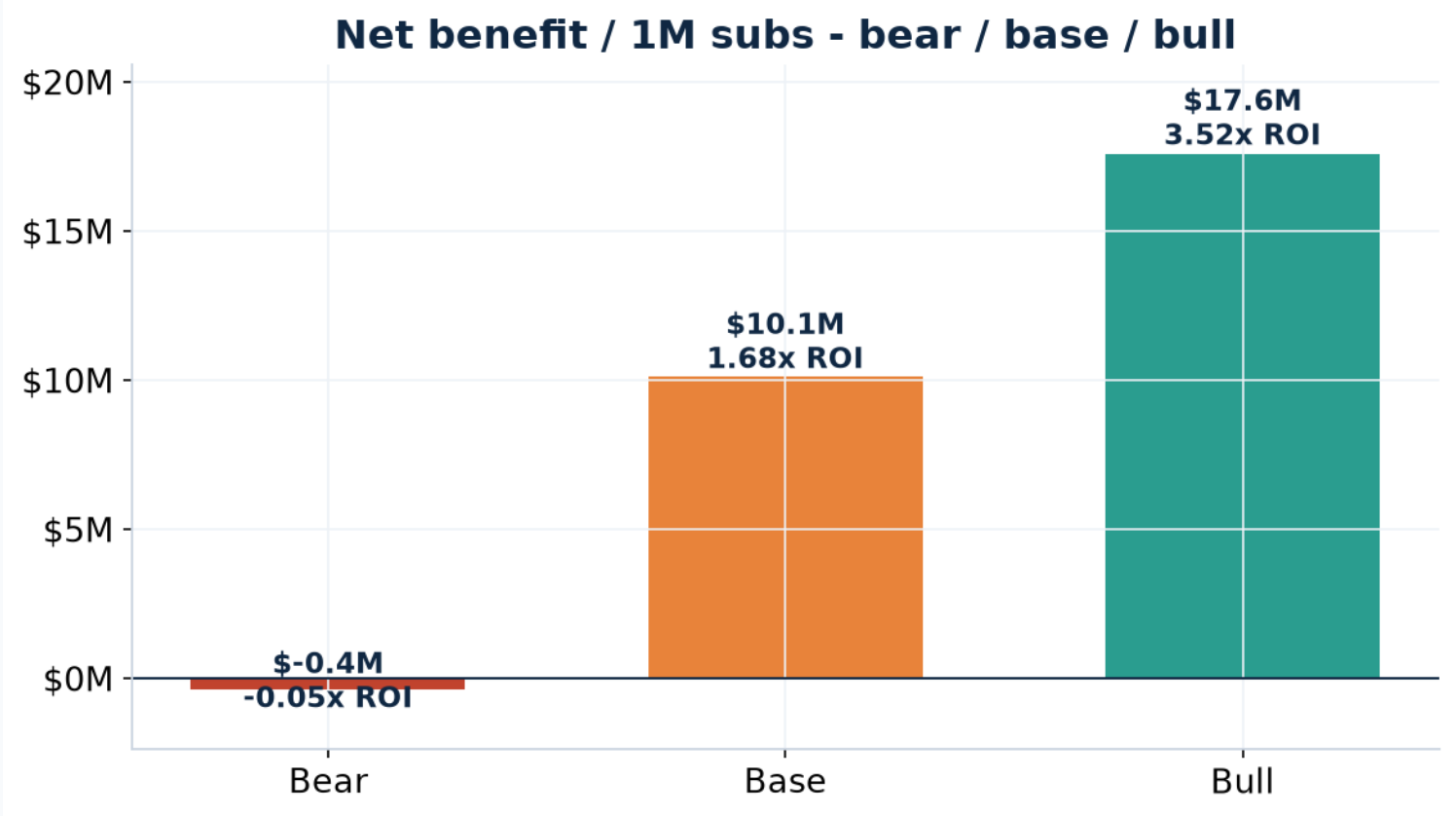
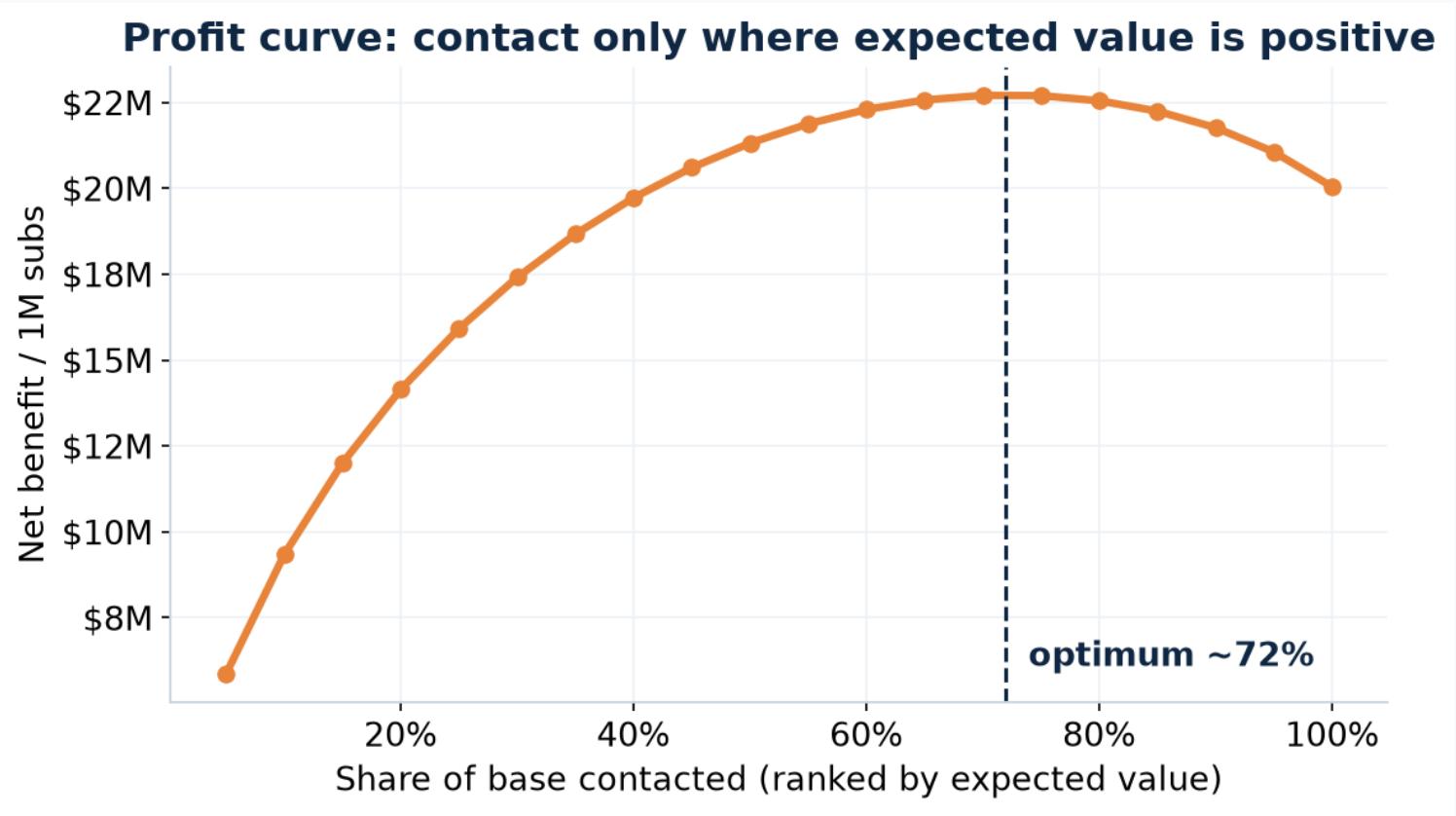
Each persona maps to a specific offer with an explicit cost, an expected save-rate, and a margin guardrail - so spend is disciplined and never margin-negative.

Persona	Problem	Offer	Cost	Save rate	Guardrail
Upgrade-Ready Switchers	Aging device triggers shopping	Upgrade / trade-in	\$50-120	Medium	Only high-EVaR
Bill-Shock Frustrated	Overage frustration	Plan right-size + autopay	\$10-30	High	No margin-negative plan
Low-Engagement Flight Risks	Usage collapse, drifting away	Loyalty call / perk	\$15-40	Low-Med	Test vs holdout
High-Value Protect-and-Hold	Valuable but at-risk	Light loyalty + concierge	\$5-15	Low	Avoid over-discount

Every offer is gated by Expected Value-at-Risk and validated against a holdout - we only spend where the expected saved margin exceeds the offer cost.

Offer costs are planning ranges to be calibrated in the pilot; save-rates validated by A/B holdout.

The business case: strong base/bull upside, bear flags the floor



\$10.1M

Net benefit / yr

1.68x

Return on spend

16,909

Customers saved / yr

\$17.6M

Bull-case upside

Base case: 22% annual churn, 50% margin, 30% save-rate, \$30 offer, \$350 CAC. Bear case (-0.05x) flags the risk; the A/B pilot resolves which scenario is real before scaling.

Roadmap: from churn prediction to causal uplift

Today we target who is most valuable AND likely to churn. The next frontier is targeting only the persuadable - customers whose behaviour the offer actually changes - using uplift (causal) modelling.

Now · Risk x Value

EVaR targeting on calibrated churn probability x customer value. Live in month 1.

LIVE

Next · Uplift modelling

Two-model / meta-learner on pilot A/B data to estimate per-customer treatment effect - stop spending on 'sure things' and 'lost causes'.

Q2

Then · Offer optimisation

Match each persuadable customer to the cheapest offer that retains them; multi-armed bandits to learn online.

Q3

Future · Real-time triggers

Stream events (dropped calls, bill spikes, usage cliffs) into live scoring for in-the-moment intervention.

Q4

Why it matters: uplift modelling typically cuts wasted retention spend materially by ignoring customers who would have stayed anyway - turning a good ROI into a great one.

Uplift modelling requires experimental (A/B) data - which is exactly what the pilot produces.

Our proposal & quotation

We propose a 12-week build of the retention engine, then an ongoing managed service. Our fee is a small fraction of the value we protect.

Phase 1 · Build (wk 1-6)

Data pipeline, model, calibration, EVaR scoring, persona logic

Phase 2 · Pilot (wk 7-12)

A/B holdout on top-20% pool; validate incremental saves & ROI

Phase 3 · Scale (ongoing)

Monthly scoring, offer optimisation, drift monitoring, QBRs

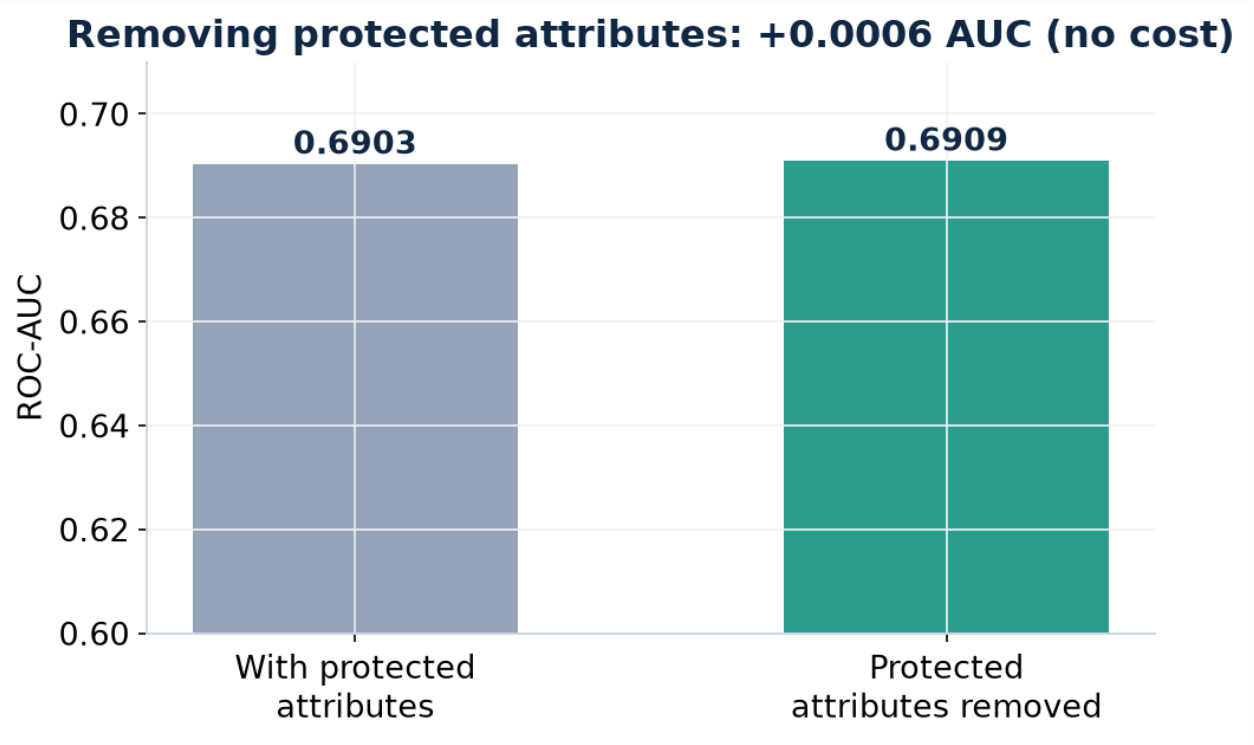
Investment

One-time setup & build	\$300k
Managed service / month	\$45k
Total year 1	\$840k

= just 8.3% of the \$10.1M net value we create

Company A keeps ~\$9.3M / yr per 1M subscribers

Fee illustrative for a PoC engagement; scales with subscriber base. Value figures per 1M subscribers / yr.



AUC 0.6903 -> 0.6909 (+0.0006, no cost)

AI-use disclosure: generative AI assisted with code scaffolding, drafting and visualisation. All analysis, numbers and conclusions were computed from the dataset and verified by the author. Every figure is reproduced by the accompanying notebook.

Fairness by design

We exclude ethnicity, marital status, income & credit class. Removing them does NOT hurt performance - AUC moves 0.6903 -> 0.6909 (+0.0006), a negligible change.

Validate before scaling

The 50/50 sample inflates apparent churn; the A/B pilot calibrates real save-rates and ROI - and tells us whether we are in the bear or bull scenario.

Monitor for drift

Re-train on fresh data and watch feature drift; a stale model silently loses accuracy as customer behaviour shifts.

Don't over-discount

Profit-curve guardrails and holdouts prevent giving margin away to customers who would have stayed anyway.

Selected references

- Reichheld & Sasser, 'Zero Defections', HBR. · Bain & Company retention research.
- Verizon / T-Mobile investor reports 2024-25. · CTIA Wireless Survey 2025.
- Verbeke et al. (2012), profit-driven churn, EJOR. · Lundberg & Lee (2017), SHAP, NeurIPS.
- scikit-learn; XGBoost; LightGBM documentation.